BLOCKCHAIN TECHNOLOGY

ASSIGNMENT 2

# CivicFix

A Decentralized Civic Issue Reporting and Tracking DApp

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| **GitHub Repository** | https://github.com/rohanshenoy30/CivicFix |

## 1. Project Overview

CivicFix is a blockchain-based decentralized application (DApp) that enables citizens to report public infrastructure issues and track their status through a transparent on-chain record. Examples include potholes, damaged roads, and broken streetlights. The application stores complaint details and status updates in a Solidity smart contract, making the report history retrievable from the blockchain.

## 2. Objectives

• Provide a simple interface for submitting civic complaints.

• Record complaint details, timestamps, reporter addresses, and status history on-chain.

• Allow users to retrieve and view reports from the deployed smart contract.

• Restrict complaint status updates to the administrator account that deployed the contract.

• Demonstrate blockchain transactions through MetaMask on a local Ganache development network.

## 3. Technology Stack

| Component | Technology |
|---|---|
| Smart contract | Solidity |
| Development and deployment | Remix IDE and Ganache |
| Wallet and transaction approval | MetaMask |
| Frontend | HTML, CSS, JavaScript |
| Blockchain interaction | ethers.js |

## 4. Smart Contract Functionality

The CivicFix smart contract defines complaint records and a status lifecycle with the states Submitted, InProgress, Resolved, and Rejected. The submitComplaint function validates the category, description, and

location before creating a complaint. The updateComplaintStatus function is restricted to the administrator and records status changes with an optional note. Read functions expose complaint details, complaint identifiers, and status history to the frontend.

## 5. Frontend and User Workflow

Users connect their MetaMask wallet, enter an issue category, location, and description, and submit the report as a blockchain transaction. The application displays recent reports, report counts, current statuses, and complaint history. The administrator desk provides a status-update workflow for the authorized deploying account. MetaMask is used to review and approve transactions on the local Ganache network.

## 6. Testing and Validation

The project testing checklist covers successful complaint submission, retrieval of stored complaint data, display of reports in the frontend, status-history retrieval, input validation for invalid complaint fields, administrator-only status updates, and rejection of invalid status transitions. Successful complaint submissions and on-chain report display are evidenced by the screenshots included in the GitHub repository. Test results should correspond to the cases actually executed in the local environment.

## 7. Deployment Environment

The smart contract was compiled in Remix IDE using the Paris EVM target for compatibility with the local Ganache network, then deployed using MetaMask. The frontend was served locally at `http://localhost:8000`. Ganache is a local development blockchain; its ETH balances are test funds and do not represent real cryptocurrency.

## 8. Repository and Evidence

The source code, README, and screenshots are available in the GitHub repository: https://github.com/rohanshenoy30/CivicFix. The repository includes evidence of the frontend home page, MetaMask transaction request, successful complaint submission, contract interaction in Remix, complaint history, and the dashboard showing multiple reports and a status update.

## 9. Conclusion

CivicFix demonstrates how a Solidity smart contract can support transparent civic complaint reporting and status tracking. By combining a browser-based interface, MetaMask transaction approvals, and a local Ethereum-compatible development network, the DApp illustrates core blockchain concepts including transactions, smart-contract validation, access control, and on-chain data retrieval.

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